

ϕ TYPES OF MATTER ϕ

→ There are in general 4 types of matter, they are:-

- Matter
- Antimatter
- Bosons
- Dark matter and Dark energy

→ Description:-

⇒ MATTER:-

- Anything that occupy space and has mass is called matter
- It is made up of proton, neutron and electron, i.e. p^+ , n^0 , e^- .

⇒ ANTIMATTER:-

- It is defined as the matter composed of particles which are “anti” of the corresponding particles of ordinary matter.
- It is made up of Antiproton (p^-), Antineutron (n^0) and Positron (e^+).

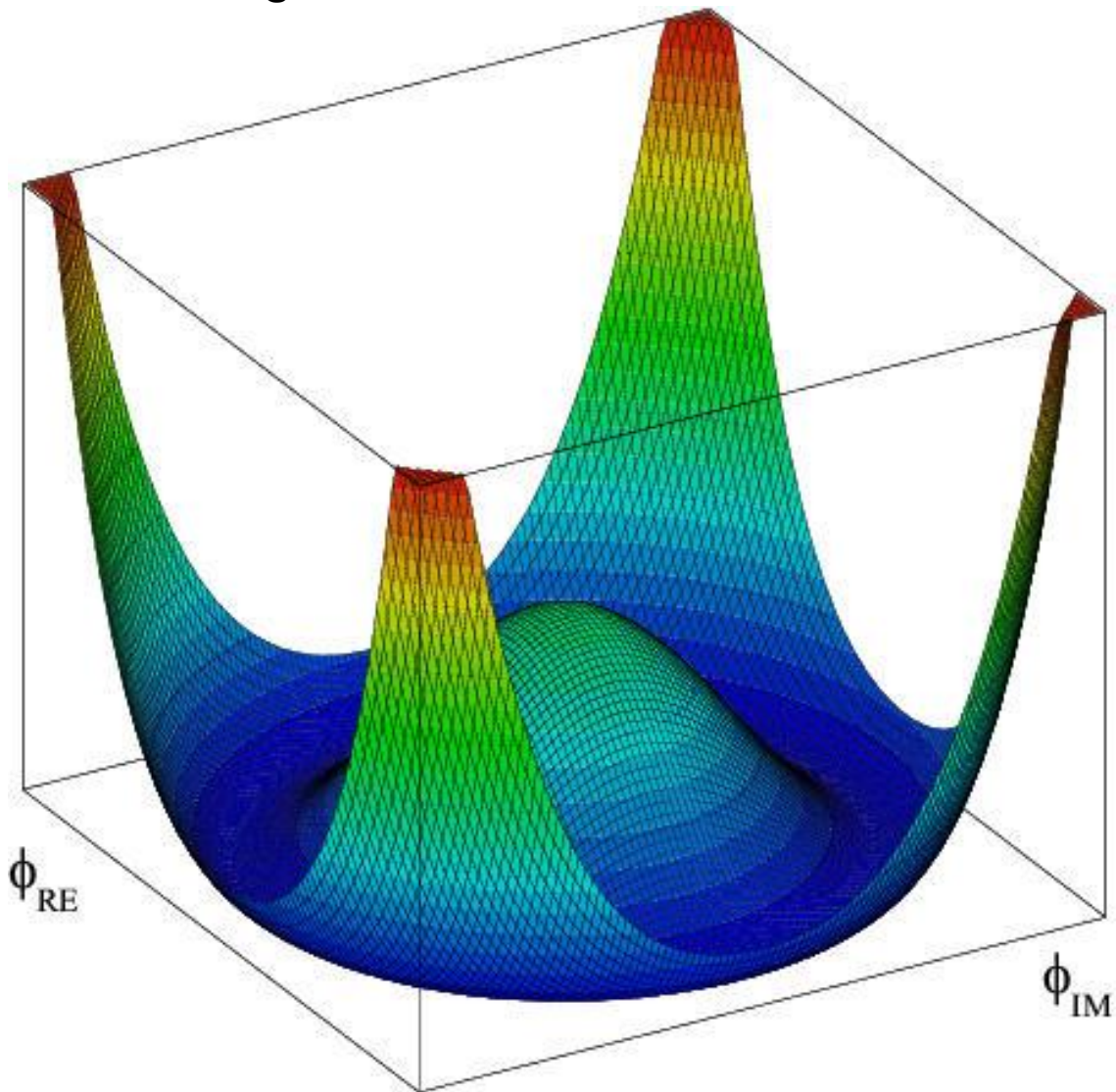
⇒ BOSONS:-

- Bosons are classified into two groups; Gauge bosons and Higgs bosons.
- Forces are carried out by gauge bosons.
- Higgs bosons are the god particles.
- Bosons don't obey the exclusion principle.
- Higgs field is a field of energy that is thought to exist in every region of the universe.
- The field is accompanied by a fundamental particle known as the higgs bosons.
- The higgs particle is a massive scalar bosons with zero spin, positive parity, no electric charge and no color change when interact with mass.
- It is also very unstable.
- Giving mass to an object is referred to as the higgs effect.
- This effect adds mass to any particles that interacts with the field.
- Gauge bosons contains following forces:
 - 1) Gravitational force [Graviton]
 - 2) Strong nuclear force [Gluon]

3) Weak nuclear force [weak bosons]

4) Electromagnetic force [Photon]

→ Below is the image giving information about how higgs field interact and give mass.



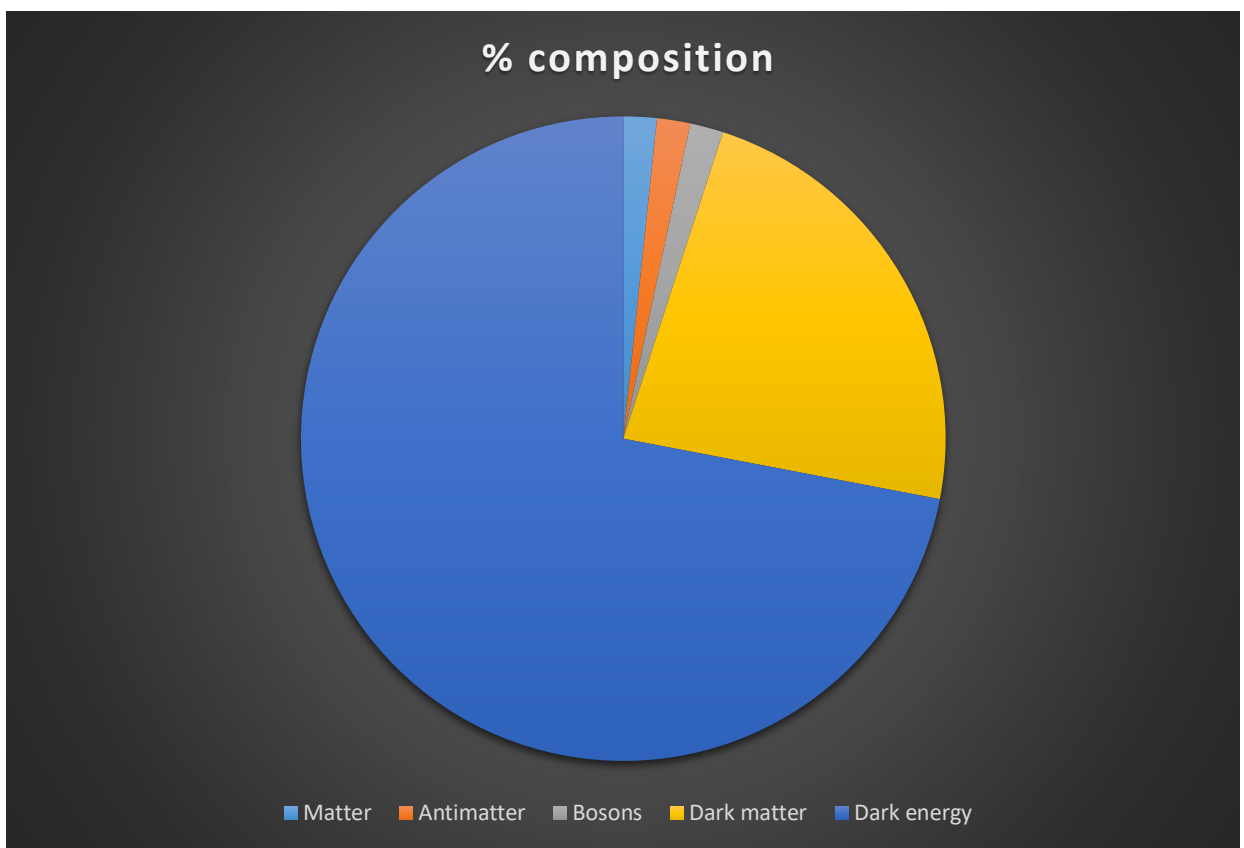
⇒ DARK MATTER & DARK ENERGY:-

- It is a hypothetical form of matter thought to account for approx. 95% in total in universe.
- Dark energy is the name given to the force that is believed to accelerate the expansion of universe.
- If there was something in empty space providing a repulsive force that makes the universe expand.
- This has been named dark energy.
- Dark matter does not emit, absorb or reflect light.



→ **Conclusion:-**

- According to recent researches, we concluded that the entire universe is not made up of only matter.
- But the percentage of the matter in the universe is less than 5%.
- Below is the pie chart representing the % composition of all 4 types of matter,



- Here, yellow is dark matter which has composition of 23%.
- Blue is dark energy has composition 72% which adds up to 95%.
- Blue, red and gray are matter, antimatter and bosons respectively, have composition 1.67% each which adds up to 5%.

THANK YOU